Homework 8

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Physics 15
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Question 1. Consider a particle in a box of size L described initially by a wave function that is a superposition of the ground and first excited states:

$$\Psi(x,0) = C[\psi_1(x) + \psi_2(x)]$$

where C is a constant and

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

are the wavefunctions for each state ($n = 1$ being the ground state, and $n = 2$ being the first excited state).

(a) Show that the value $C = 1/\sqrt{2}$ normalizes this wave function, assuming ψ_1 and ψ_2 are themselves normalized.

(b) Find $\Psi(x,t)$ at any later time t.

(c) Show that the superposition is not a stationary state, but that the average energy in this state is the arithmetic mean $(E_1 + E_2)/2$ of the ground and first excited-state energies E_1 and E_2.

Solution. For part (a):

$$\begin{aligned}
|\Psi|^2 &= \int_0^L C^2(\psi_1^2 + 2\psi_1\psi_2 + \psi_2^2)dx \\
&= \frac{1}{2}\left[1 + 2\int_0^L \left(\sqrt{\frac{2}{L}}\sin\left(\frac{1\pi x}{L}\right)\right)\left(\sqrt{\frac{2}{L}}\sin\left(\frac{2\pi x}{L}\right)\right)dx + 1\right] \\
&= 1 + \int_0^L \frac{2}{L}\sin\left(\frac{1\pi x}{L}\right) 2\sin\left(\frac{\pi x}{L}\right)\cos\left(\frac{\pi x}{L}\right)dx \\
&= 1 + \frac{4}{L}\int_0^L \sin^2\left(\frac{\pi x}{L}\right)\cos\left(\frac{\pi x}{L}\right)dx.
\end{aligned}$$

It is trivial that the function within the integral is an odd function when centered at $x = \frac{L}{2}$. Therefore, the integral evaluates to zero, and

$$\boxed{|\Psi|^2 = 1.}$$

For part (b), the ground and first excited wavefunctions can be expressed with time-dependence as

$$\Psi_1 = \psi_1(x)e^{-iE_1t/\hbar}, \Psi_2 = \psi_2(x)e^{-iE_2t/\hbar},$$

where E_1 and E_2 are the energy states of the wavefunctions respectively. Since

$$E = \frac{p^2}{2m} = \frac{\hbar^2k^2}{2m} = \frac{\hbar^2\pi^2n^2}{2mL^2},$$

therefore

$$E_1 = \frac{\hbar^2\pi^2}{2mL^2}, E_2 = 4E_1 = \frac{2\hbar^2\pi^2}{mL^2}.$$

Therefore,

$$\begin{aligned}\Psi(x,t) = C[\Psi_1 + \Psi_2] &= \frac{1}{\sqrt{2}}\left[\sqrt{\frac{2}{L}}\sin\left(\frac{\pi x}{L}\right)e^{-iE_1t/\hbar} + \sqrt{\frac{2}{L}}\sin\left(\frac{2\pi x}{L}\right)e^{-iE_2t/\hbar}\right] \\ &= \boxed{\frac{1}{\sqrt{L}}\left[\sin\left(\frac{\pi x}{L}\right)e^{-i\hbar\pi^2t/2mL^2} + \sin\left(\frac{2\pi x}{L}\right)e^{-2i\hbar\pi^2t/mL^2}\right].}\end{aligned}$$

For part (c), note that $\Psi(x,t) = c_1\Psi_1 + c_2\Psi_2$. The average energy is

$$\begin{aligned}\langle E\rangle &= \int_{-\infty}^{\infty}(c_1^*\Psi_1^* + c_2^*\Psi_2^*)\hat{H}(c_1\Psi_1 + c_2\Psi_2)dx \\ &= \int_{-\infty}^{\infty}(c_1^*\Psi_1^* + c_2^*\Psi_2^*)(c_1E_1\Psi_1 + c_2E_2\Psi_2)dx = \int_{-\infty}^{\infty} c_1^*c_1\Psi_1^*\Psi_1E_1 + \cdots + c_2^*c_2\Psi_2^*\Psi_2E_2dx.\end{aligned}$$

The middle cross terms are zero because the two eigenstates are orthogonal.

$$\begin{aligned}\langle E\rangle &= |c_1|^2E_1\int_{-\infty}^{\infty}|\Psi_1|^2dx + |c_2|^2E_2\int_{-\infty}^{\infty}|\Psi_2|^2dx = \left(\frac{1}{\sqrt{2}}\right)^2E_1(1) + \left(\frac{1}{\sqrt{2}}\right)^2E_2(1) \\ &= \boxed{\frac{E_1 + E_2}{2}.}\end{aligned}$$

□

Question 2a. (Eisberg & Resnick 6.12) Verify by substitution that the exponential general solutions, (6-63) and (6-64), satisfy the time-independent Schroedinger equation (6-13) for the finite square well potential in the regions outside the well.

Solution. Recall that the time-independent Schroedinger equation (6-13) is

$$-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi(x) + V\psi(x) = E\psi(x).$$

Let us first substitute (6-63)

$$\psi(x) = Ce^{\kappa x} + De^{-\kappa x}, \quad \kappa = \frac{\sqrt{2m(V-E)}}{\hbar}.$$

The 2nd partial of the wavefunction is

$$\frac{\partial^2}{\partial x^2}\psi(x) = \frac{\partial}{\partial x}\kappa(Ce^{\kappa x} - De^{\kappa x}) = \kappa^2(Ce^{\kappa x} + De^{\kappa x}) = \kappa^2\psi(x).$$

Then,

$$\begin{aligned}
-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi(x) + V\psi(x) &= -\frac{\hbar^2}{2m}\kappa^2\psi(x) + V\psi(x) \\
&= -\frac{\hbar^2}{2m}\left(\frac{\sqrt{2m(V-E)}}{\hbar}\right)^2\psi(x) + V\psi(x) \\
&= -(V-E)\psi(x) + V\psi(x) = \boxed{E\psi(x).}
\end{aligned}$$

Without loss of generality, the same procedure applies to equation (6-64). □

Question 2b. (Eisberg & Resnick 6.31) The restoring force constant C for the vibrations of the interatomic spacing of a typical diatomic molecule is about 10^3 joules/m^2. The mass of the molecule is 4.1×10^{-26} kg.

(a) Estimate the difference in energy between the ground state and first excited state of this vibrating molecule.

(b) From this estimate determine the energy of the photon emitted by the vibrations in the charge distribution when the system makes a transition between the first excited state and the ground state.

(c) Determine also the frequency of the photon, and compare it with the classical oscillation frequency of the system.

(d) In what range of the electromagnetic spectrum is it?

Solution. Recall that the difference between neighboring energy states is always $\Delta E = \hbar\omega$. Hence, we need to find ω. If we treat this system as a quantum harmonic oscillator, we realize that the angular frequency is

$$\omega = \sqrt{\frac{C}{m}} = \sqrt{\frac{10^3 \text{ J/m}^2}{4.1 \times 10^{-26} \text{ kg}}} \approx 1.56 \times 10^{14} \text{ rad/s}.$$

Therefore,

$$\Delta E = \hbar\omega = \frac{6.63 \times 10^{-34} \text{ J} \cdot \text{s}}{2\pi}(1.56 \times 10^{14} \text{ rad/s}) = \boxed{1.65 \times 10^{-20} \text{ J.}}$$

This immediately answers part (b), because the energy of the photon emitted is simply the change in the energy levels. For part (c), we simply find frequency by

$$f = \frac{\omega}{2\pi} = \frac{1.56 \times 10^{14} \text{ rad/s}}{2\pi} = \boxed{2.48 \times 10^{13} \text{ Hz.}}$$

In response to part (d), this frequency corresponds to $\boxed{\text{infrared light.}}$ □

Question 3. Particle in an atom-sized box.

(a) Using the results of the "particle in a box" that we derived in class, find an expression for the energy levels of an electron in a one-dimensional box that is roughly the size of an atom. Express your answer in units of electron volts and the quantum number "n".

(b) Using your expression from part (a), calculate the energy of the ground state in electron volts.

(c) How does this energy compare with the ground state energy of Hydrogen. Comment on the result.

(d) What is the wavelength of a photon emitted when the electron makes a transition from the first excited state to the ground state?

(e) Now consider a box with $L = 1$ cm, and an electron with roughly the same energy you calculated in part (b). In what range would the n-value for the electron in the box lie? Comment on this result (Hint: What are the energy level spacings for adjacent values of n near this value and how does this compare with the classical treatment of a macroscopic box?)

Solution. Recall that for lengths of atoms, $L \sim 1\text{Å} = 10^{-10}$ m. Hence, for part (a),

$$E = \frac{n^2\hbar^2\pi^2}{2mL^2} = \frac{n^2(6.63 \times 10^{-34}\ \text{J} \cdot \text{s})^2}{8(9.11 \times 10^{-31}\ \text{kg})(10^{-10}\ \text{m})^2}\left(\frac{1\ \text{eV}}{1.6 \times 10^{-19}\ \text{J}}\right) \approx \boxed{37.70n^2\ \text{eV.}}$$

Hence, for part (b) the ground state is simply $\boxed{E_1 = 37.70\ \text{eV.}}$ For part (c), recall that the ground state energy of a Hydrogen atom is 13.6 eV, meaning that this energy is almost three times greater than the Hydrogen ground state energy.

For part (d), the energy emitted from the first excited state to the ground state is

$$\Delta E = E_2 - E_1 = [(2)^2 - (1)^2](37.7\ \text{eV}) \approx \boxed{113\ \text{eV.}}$$

Therefore, the wavelength of the photon emitted is

$$\lambda = \frac{\hbar}{E} = \frac{1240\ \text{eV} \cdot \text{nm}}{113\ \text{eV}} \approx \boxed{11\ \text{nm}}.$$

For part (e), for equivalent energies with differing lengths and principal quantum numbers, we get the relationship that

$$\frac{n_1^2\hbar^2\pi^2}{2mL_1^2} = \frac{n_2^2\hbar^2\pi^2}{2mL_2^2},$$

in other words,

$$\frac{n_2^2}{n_1^2} = \frac{L_2^2}{L_1^2} \implies \frac{n_2}{n_1} = \frac{L_2}{L_1}.$$

Hence, if we scale the box to a length $L_2 = 1$ cm, then

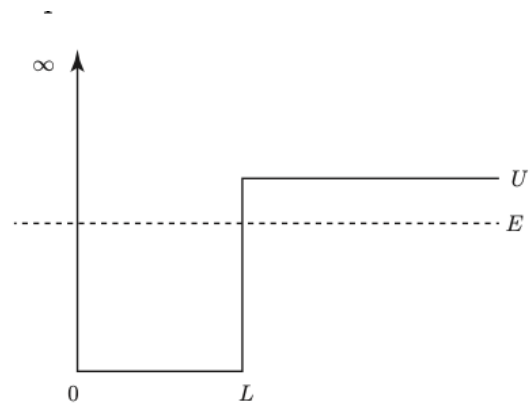
$$\frac{L_2}{L_1} = \frac{10^{-2}\ \text{m}}{10^{-10}\ \text{m}} = 10^8.$$

Then,

$$n_2 = 10^8(n_1) = 10^8(1) = \boxed{10^8.}$$

This is a very big quantum number, and it is proof of the Correspondence Principle. Since higher quantum numbers approach a continuous energy spectrum, the system is shifting towards classical mechanics instead of quantum mechanics. $\square$

Question 4. (Serway 6.23) Consider a square well having an infinite wall at $x = 0$ and a wall of height U at $x = L$ (see figure). For the case $E < U$, obtain solutions to the Schrodinger equation inside the well ($0 \leq x \leq L$) and in the region beyond ($x > L$) that satisfy the appropriate boundary conditions at $x = 0$ and $x = \infty$. Enforce the proper matching conditions at $x = L$ to find an equation for the allowed energies of this system. Are there conditions for which no solution is possible? Explain.



Solution. To solve this we must conceptually understand this type of well. Because the left wall of the well is an infinite wall, the potential must be zero when $x \leq 0$. However, because the right wall of the well is a finite well, we know that there must be some sort of exponential decay. Within the well, there is some oscillation. Hence, we are given two equations as such

$$\psi_{\mathrm{I}}(x) = Ae^{ikx} + Be^{-ikx}, \quad k = \frac{\sqrt{2mE}}{\hbar}$$
$$\psi_{\mathrm{II}}(x) = Ce^{-\kappa x}, \quad \kappa = \frac{\sqrt{2m(U-E)}}{\hbar}.$$

Now let us solve for the boundary conditions. Firstly, since the function is continuous, it must be true that

$$\psi_{\mathrm{I}}(0) = Ae^0 + Be^0 = A + B = 0,$$

implying that $B = -A$ and therefore,

$$\psi_I(x) = Ae^{ikx} - Ae^{ikx} = A(e^{ikx} - e^{-ikx}).$$

By the same logic, it is true that

$$\psi_{\mathrm{I}}(L) = \psi_{\mathrm{II}}(L) \implies \boxed{A(e^{ikL} - e^{-ikL}) = Ce^{-\kappa L}}, \tag{1}$$

and

$$\psi'_{\mathrm{I}}(L) = \psi'_{\mathrm{II}}(L) \implies \boxed{iAk(e^{ikL} + e^{-ikL}) = -\kappa Ce^{-\kappa L}}. \tag{2}$$

Dividing (2) by (1) gives us

$$ik\left(\frac{e^{ikL}+e^{-ikL}}{e^{ikL}-e^{-ikL}}\right)=-\kappa.$$

By rules of complex exponentials, where $e^{ikx}+e^{-ikx}=2\cos(kx)$ and $e^{ikx}-e^{-ikx}=2i\cos(kx)$, this equation reduces to

$$k\cot(kL)=-\kappa, \tag{3}$$

which is a transcendental equation. To solve this equation, we have to computationally find it. This is easier when we make the equation dimensionless. Let us define a baseline energy $\mathcal{E}_0=\frac{\hbar^2}{2mL^2}$, and make a note that

$$k^2+\kappa^2=\frac{2mE+2m(U-E)}{\hbar^2}=\frac{2mU}{\hbar^2}. \tag{4}$$

We define

$$\boxed{\xi=kL=\sqrt{\frac{2mL^2E}{\hbar^2}}=\sqrt{\frac{E}{\mathcal{E}_0}}.}$$

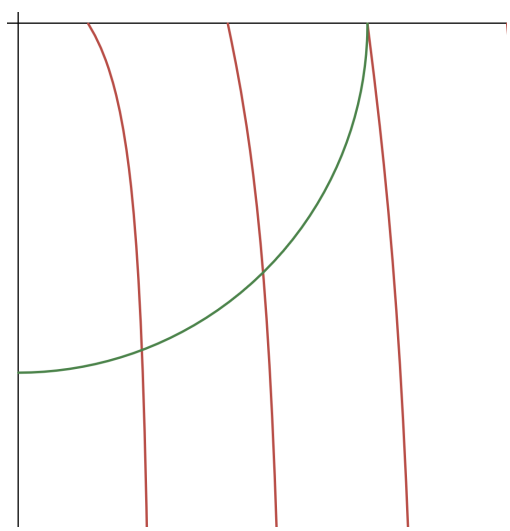
Next, by multiplying both sides of (4) by L^2, we define

$$(kL)^2+(\kappa L)^2=\xi^2+(\kappa L)^2=\boxed{\eta^2=\frac{2mUL^2}{\hbar^2}=\frac{U}{\mathcal{E}_0}.}$$

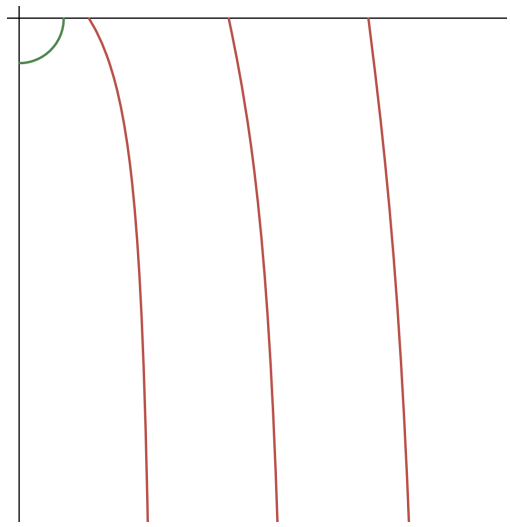
With this, we can make equation (3) dimensionless by multiplying both sides by L to get

$$kL\cot(kL)=-\kappa L \implies \boxed{\xi\cot(\xi)=-\sqrt{\eta^2-\xi^2}.}$$

Now, let us plot this equation as two lines with ξ on the x-axis as such:



From the figure, each intersection represents a solution, and by analysis I realized that the number of solutions $n\in\mathbb{N}$ is based on the smallest possible n such that $\eta\leq\left(n-\frac{1}{2}\right)\pi$. Hence, it is possible to have no solutions when $\boxed{\eta<\frac{\pi}{2}}$ as shown by the figure below.



This is achievable if we either *reduce the well depth* U or *narrow the well length* L to a certain degree. $\square$

Question 5. A particle of mass m moves in a three-dimensional box with edge lengths L_1, L_2, and L_3. Find the energies of the six lowest states if $L_1 = L, L_2 = 2L$, and $L_3 = 2L$. Which of these energies are degenerate?

Solution. Let us define the box to be a $L \times 2L \times 2L$ on the x, y, and z-axes, respectively. Given an infinite 3D well, the three wavefunctions components are

$$X(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n_x \pi x}{L}\right),$$
$$Y(y) = \sqrt{\frac{1}{L}} \sin\left(\frac{n_y \pi y}{2L}\right),$$
$$Z(z) = \sqrt{\frac{1}{L}} \sin\left(\frac{n_z \pi z}{2L}\right),$$

where, by separation of variables, $\psi(x) = X(x)Y(y)Z(z)$. Therefore, the total energy is

$$E(n_x, n_y, n_z) = E_x + E_y + E_z = \frac{n_x^2 \pi^2 \hbar^2}{2mL^2} + \frac{n_y^2 \pi^2 \hbar^2}{2m(2L)^2} + \frac{n_z^2 \pi^2 \hbar^2}{2m(2L)^2} = \frac{\pi^2 \hbar^2}{8mL^2}(4n_x^2 + n_y^2 + n_z^2).$$

Now to find the lowest energy states, we must find use the smallest integers in n_x, n_y, n_z such that E is minimized. Since increasing n_x increases E at a greater rate than increasing n_y or n_z, let us incrementally increase n_y and n_z before increasing n_x:

$$E(1,1,1) = \frac{\pi^2 \hbar^2}{8mL^2}(4(1)^2 + (1)^2 + (1)^2) = \boxed{\frac{3\pi\hbar^2}{4mL^2}},$$
$$E(1,1,2) = \frac{\pi^2 \hbar^2}{8mL^2}(4(1)^2 + (1)^2 + (2)^2) = \boxed{\frac{9\pi\hbar^2}{8mL^2}},$$
$$E(1,2,1) = \frac{\pi^2 \hbar^2}{8mL^2}(4(1)^2 + (2)^2 + (1)^2) = \frac{9\pi\hbar^2}{8mL^2},$$
$$E(1,2,2) = \frac{\pi^2 \hbar^2}{8mL^2}(4(1)^2 + (2)^2 + (2)^2) = \boxed{\frac{3\pi\hbar^2}{2mL^2}},$$
$$E(1,1,3) = \frac{\pi^2 \hbar^2}{8mL^2}(4(1)^2 + (1)^2 + (3)^2) = \boxed{\frac{7\pi\hbar^2}{4mL^2}},$$
$$E(1,3,1) = \frac{\pi^2 \hbar^2}{8mL^2}(4(1)^2 + (1)^2 + (3)^2) = \frac{7\pi\hbar^2}{4mL^2},$$
$$E(1,2,3) = \frac{\pi^2 \hbar^2}{8mL^2}(4(1)^2 + (2)^2 + (3)^2) = \boxed{\frac{17\pi\hbar^2}{8mL^2}},$$
$$E(1,3,2) = \frac{\pi^2 \hbar^2}{8mL^2}(4(1)^2 + (3)^2 + (2)^2) = \frac{17\pi\hbar^2}{8mL^2},$$
$$E(2,1,1) = \frac{\pi^2 \hbar^2}{8mL^2}(4(2)^2 + (1)^2 + (1)^2) = \boxed{\frac{9\pi\hbar^2}{4mL^2}}.$$

Notice that $E(1,1,2) = E(1,2,1)$, $E(1,1,3) = E(1,3,1)$, and $E(1,2,3) = E(1,3,2)$. Hence, the energies $\boxed{E = \frac{9\pi\hbar^2}{8mL^2}, \frac{7\pi\hbar^2}{4mL^2}, \frac{17\pi\hbar^2}{8mL^2}}$ are all degenerate. $\square$